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$$1) \lim_{x \rightarrow +\infty} \frac{e^{2x} + 2e^x}{e^x + 3e^{2x}} = \frac{\cancel{e^{2x}} (1 + e^{-x})}{\cancel{e^{2x}} (3 + e^{-x})} = \frac{1}{3}$$

$$2) \lim_{x \rightarrow +\infty} \left(\frac{x^2 - x}{x^2 - 1} \right)^{3x} = \left(\frac{x \cancel{(x-1)}}{\cancel{(x-1)}(x+1)} \right)^{3x} = \left(\frac{x}{x+1} \right)^{3x} = \left(\frac{x+1}{x} \right)^{-3x} = \left(1 + \frac{1}{x} \right)^{-3x} = e^{-3}$$

$$3) \lim_{x \rightarrow +\infty} \sin \left[\log(\sqrt{x} + 1) - \log(\sqrt{x-1}) \right] = \sin \left[\log(\sqrt{x} (1 + \frac{1}{\sqrt{x}})) - \log(\sqrt{x} (1 + \frac{1}{x})) \right] = \log(\sqrt{x}) + \log\left(1 + \frac{1}{\sqrt{x}}\right) - \log(\sqrt{x}) - \log\left(1 + \frac{1}{x}\right)$$

$$\sin \left[\log\left(1 + \frac{1}{\sqrt{x}}\right) - \frac{1}{2} \log\left(1 + \frac{1}{x}\right) \right] = \sin \left[\frac{1}{\sqrt{x}} + o\left(\frac{1}{\sqrt{x}}\right) - \frac{1}{2x} + o\left(\frac{1}{x}\right) \right] = \sin \left(\frac{1}{\sqrt{x}} + o\left(\frac{1}{\sqrt{x}}\right) \right)$$

$$\frac{\sin x}{\sqrt{x}} (1 + o(1)) \rightarrow 0 \text{ per } x \rightarrow +\infty$$

$$4) \lim_{n \rightarrow +\infty} n \sin\left(\frac{1}{n}\right) \cos\left(\frac{1}{n}\right) + o(\tan n) = n \left(\left(\frac{1}{n} + o\left(\frac{1}{n}\right) \right) \left(1 - \frac{1}{2n^2} + o\left(\frac{1}{n^2}\right) \right) \right)$$

$$(1 + o(1)) \left(1 - \frac{1}{2n^2} + o\left(\frac{1}{n^2}\right) \right) + o(\tan n) = 1 + \frac{\pi}{2}$$

$$5) \lim_{x \rightarrow 0^+} \left(\frac{1}{\sin x} \right)^{-\frac{1}{\sin x}} = e^{-\frac{1}{\sin x} \cdot \ln(1/\sin x)} \quad y = \frac{1}{\sin x}$$

$$\lim_{y \rightarrow +\infty} e^{-y} y = 0$$

$$6) \lim_{x \rightarrow 3^-} (x^2 - 9) \sqrt{\frac{2x^3}{3-x}} = (x+3)(x-3) \frac{\sqrt{2x^3}}{\sqrt{3-x}} = -(x+3)(3-x) \frac{\sqrt{2x^3}}{\sqrt{3-x}} = (x+3)\sqrt{3-x} \sqrt{2x^3} = 0$$

$$7) \lim_{x \rightarrow 0^+} \frac{\sqrt{x} + \sqrt{1-x} - 2}{x^\alpha}, \alpha > 0$$

$$\frac{1 - \frac{x}{2} + o(x) + 1 - \frac{x}{2} + o(x) - 2}{x^\alpha} = \frac{-x + o(x)}{x^\alpha} = \frac{x(-1 + o(1))}{x^\alpha} = \frac{-1 + o(1)}{x^{\alpha-1}}$$

$$8) \lim_{x \rightarrow 0^+} x^{\frac{-1}{x^2}} = e^{\frac{-\ln(x)}{2 \ln(x)}} = e^{-1/2}$$

$$9) \lim_{x \rightarrow \pm \infty} \left(\frac{x^2 - 2x}{x^2 + 1} \right)^{3x} = \left(\frac{x^2 \left(1 - \frac{2}{x}\right)}{x^2 \left(1 + \frac{1}{x^2}\right)} \right)^{3x} = \left(1 - \frac{2}{x}\right)^{3x} = e^{-6}$$

$$10) \lim$$